AMA3640 Statistical Inference

2025–2026 Semester 1

Time: **Lecture**: 12:30–14:20, Mon, SHA030;
Tutorial: 17:30–18:20, Fri, Z207/Z211

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Course Schedule (Tentative; 22 August 2025):

Week	Date	Content	Remark
1	1 Sep	Method of moments; Maximum likelihood estimation	
2	8 Sep	Maximum likelihood estimation; Evaluation of point estimators	
3	15 Sep	Best unbiased estimator; Cramér–Rao lower bound	
4	22 Sep	Consistency; Asymptotic normality	
5	29 Sep	Sufficient statistic	Assignment 1
6	6 Oct	Complete statistic	
7	13 Oct	Rao–Blackwell theorem; Best unbiased estimator	
8	20 Oct	Examples for point estimation; Testing statistical hypotheses	
9	27 Oct	Testing statistical hypotheses; Most powerful tests	Assignment 2
10	3 Nov	Uniformly most powerful tests; Likelihood ratio test	
11	10 Nov	Interval estimation	Midterm test
12	17 Nov	Bayesian inference	
13	24 Nov		Assignment 3

Assessment:

Exam 60% + Midterm 20% + Assignment×3 20%

References:

- Bain, L. J. and Engelhardt, M. (2000). *Introduction to Probability and Mathematical Statistics* (2nd ed.). Duxbury.
- Casella, G. and Berger, R. L. (2002). *Statistical Inference* (2nd ed.). Duxbury.
- Hogg, R. V., McKean, J. W., and Craig, A. T. (2012). *Introduction to Mathematical Statistics* (7th ed.). Prentice Hall.